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Effect of management on the spatial spread of mountain pine beetle (*Dendroctonus ponderosae*) in Banff National Park

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ABSTRACT

To evaluate control measures, the spread of mountain pine beetles, *Dendroctonus ponderosae*, in management and monitoring regions in Banff National Park was analyzed for years 1997 to 2004. The Park allowed mountain pine beetles to follow their natural course in a monitoring zone (74,041 ha), whereas in a management zone (45,997 ha) an extensive eradication program was established in 2001 which included baiting mountain pine beetles and cutting and burning all colonized trees. We used data collected from an annual aerial survey and the geo-referenced location of trees that were cut and removed to assess if the area colonized and the spatial extent of mountain pine beetles differed between the two zones. After 4 years, the control measures did not significantly affect the area colonized by mountain pine beetles, and in 2004 mountain pine beetles were still expanding in both zones, although at a slow rate (1.23 ha per year). The spatial extent and the rate at which 500 m × 500 m cells (25 ha) were colonized were much reduced in the management zone. Thus, the management program appeared to have reduced the success of long distance movement as measured by colonizing new 25 ha cells, but short distance dispersal remained successful as indicated by the continued increase in area colonized. The effectiveness of control measures was probably limited by the number of colonized trees that were missed and by survival rates determined by other untested factors.

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1. Introduction

A major impact on ecosystems in temperate coniferous forests are bark beetles (Coleoptera: Curculionidae, Scolytinae) (Safranyik, 1995; Clarke and Billings, 2003; Wermelinger, 2004; Økland and Berryman, 2004; Wagner et al., 2006). Some species, particularly in the genus *Dendroctonus* in North America, breed preferentially in live mature trees that they subsequently kill. Management options are limited because beetles are protected under the bark of trees for most of their life cycle, preventing remotely applied insecticides. The other difficulty is that remote detection of infested trees is delayed because changes in the colouration of tree foliage only becomes clearly evident once the beetles have already reproduced, killed their natal tree and colonized new trees (Wulder et al., 2006). Consequently, identification and removal of infested trees requires labour-intensive ground surveys for direct control. One way to make these surveys more targeted is to bait trees with beetle pheromones to aggregate them in known locations that can then be checked at the end of the summer season of beetle colonization (Carroll et al., 2006). These trees are then treated, typically by felling and burning, to kill beetles prior to their emergence.

The breeding biology of bark beetles is such that high success in removing infested trees is critical for population reduction (Carroll et al., 2006). While most tree-killing species require a minimum population size to be able to mass-attack trees to overwhelm the defences of live trees (Allee effect, Raffa and Berryman, 1983), their pheromone communication system allows beetles to aggregate on individual trees. For trees with thick phloem, the number of offspring produced per parental female can be high, so that untreated trees can continue to cause a significant impact on the trees killed in subsequent years (Amman, 1972; Carroll et al., 2006).

Assessment of the success of efforts to manage bark beetles is challenging. The scale over which bark beetles occur during outbreaks (thousands of hectares) means that numerous different land tenure jurisdictions are involved that usually have different control methods and data collection protocols (Carroll et al., 2004). Further, managers are reluctant to leave some areas untreated to

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serve as experimental controls, so that it is unreasonable to expect well-replicated management experiments (although there are examples of small-scale studies particularly when novel approaches are introduced, e.g. Borden et al., 1983; Preisler and Mitchell, 1993; Borden, 1995; Bentz and Munson, 2000). To examine the effectiveness of standard control efforts, protected areas can provide a non-managed landscape with which to examine the effectiveness of management practices (Clarke and Billings, 2003). Here we examine the impact of management tactics employed by Banff National Park (Alberta, Canada) in part of its landbase to limit the increases of mountain pine beetles, *Dendroctonus ponderosae*.

Parks Canada has a mandate to maintain and allow natural disturbance such as fire, flooding and insect outbreaks as a part of the broader goal of maintaining ecological integrity (Canada National Parks Act, 2000). However, pressure from industry outside the park to stop the spread of mountain pine beetles to provincial lands led Banff to implement a large-scale 'management experiment'. In 2001, the region surrounding the Banff town site was divided into two zones, which we refer to as the management and monitoring zones. To examine whether these zones differed in terms of the increase in mountain pine beetles, we first established how the zones differed in the abundance of habitat for mountain pine beetles, and then contrasted several metrics of mountain pine beetles distribution and abundance. This study provides a rare quantitative comparison of a bark beetle management tactic at a large scale.

2. Methods

2.1. Study area and outbreak history

A brief timeline of mountain pine beetle outbreak history, data collection and management in Banff National Park is provided in Table 1. In 1998, Banff National Park personnel noticed trees colonized by mountain pine beetles a few kilometres southwest of the Banff townsite (Healy Creek), and ground surveys revealed that beetles were present in 1996 or 1997. At such an initial stage when populations are small and well defined, direct control measures are recommended (Carroll et al., 2006). The region surrounding the Banff town site was divided into two management zones in 2001 (Fig. 1). In the management zone (east of town site, 45,997 ha), trees in selected areas were baited with mountain pine beetle pheromone baits (Phero Tech Inc.) in June each year. All trees (baited or unbaited) detected with newly colonized mountain pine beetles were felled and burned in the following fall and winter. In the monitoring zone (west of townsite, 74,041 ha), natural process were largely allowed to occur. Between 2001 and 2004, 1,369 baits were deployed in the management zone with most of the effort in Tunnel Mountain and Lower Fairholme areas (Table 2, Fig. 2). In addition to baiting trees in the management zone, prescribed burns during spring (April to June) were a management tool in both zones, in part to control mountain pine beetles by

Table 1
Time line of events pertaining to mountain pine beetle population dynamics, habitat and data collection in Banff National Park

Event	Year
First outbreak of mountain pine beetle detected	1940
Approximately 15,500 trees removed	1941–1942
First outbreak over	1943
Banff-Jasper Ecological Land Classification	1974–1982
First prescribed burn, killing susceptible lodgepole pine	1983
129 trees colonized in the upper Spray Lakes (south of study area). No outbreak followed	1983
Second outbreak of mountain pine beetle detected. Start of outbreak presumed to have occurred 1 or 2 years earlier.	1998
Aerial surveys start	1998
Study area divided into two zones: monitoring and management	2001
First pheromone baits deployed	2001
First trees cut and removed	2001
Large fire (5188 ha) on Fairholme bench	2003

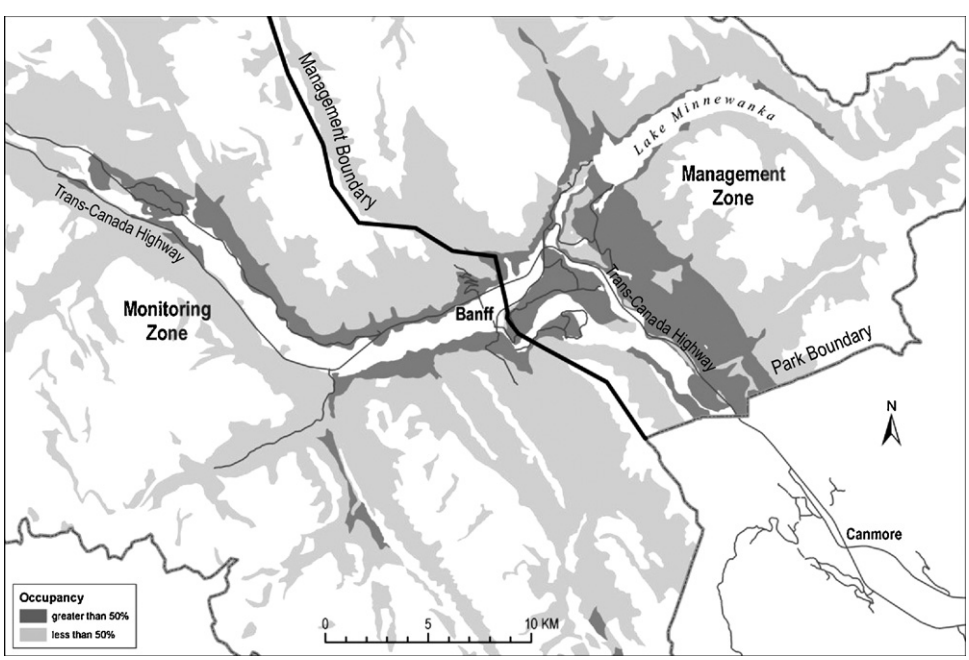

Fig. 1. Mountain pine beetle habitat in vegetation types of high occupancy (dark shading: >50%) and low occupancy (light shading: <50%) (see Table 2); white areas either do not contain lodgepole pine or lodgepole pine is <60 years old. The black line delineates the monitoring (west) and management (east) zones. Areas outside the park (to the northeast, southeast, and southwest) were not considered.

Table 2
The characteristics of the monitoring and management zones and the mountain pine beetle outbreak in Banff National Park from 1998 to 2004

Zone	Sub-zone	Total area (ha)	Habitat		Area affected		Baits deployed	Trees removed
			ha	% total	ha	% habitat		
Monitoring		74,041	32,874	44.4	218.4	0.7	0	0
	Hillsdale	5,110	3,875	70.8	62.7	1.9	0	0
	Healy Creek	4,567	3,232	75.8	35.3	0.9	0	0
	Norquay 1	840	797	94.9	120.3	15.1	0	0
	Tunnel Mt. 1	16	13	79.8	0.0	0.0	0	0
Management		45,997	23,090	50.2	383.2	1.2	1,369	4,883
	Norquay 2	114	93	82.0	6.6	7.1	0	0
	Tunnel Mt. 2	1,199	930	77.6	121.1	13.0	480	2,464
	Bumpy Meadows	2,239	2,014	90.0	63.8	3.2	225	560
	Lower Fairholme	2,025	1,542	76.1	88.4	5.7	386	901
	Upper Fairholme	2,705	2,447	90.5	103.2	4.2	278	958

Sub-zones are as defined in Fig. 2. Both the Norquay and Tunnel Mountain sub-zone span the control line, but the areas are small and management was consistent with the larger zone. Total area of habitat and area affected is larger than the sum of the sub-zones because sub-zones do not cover the entire zone.

killing susceptible lodgepole pine. During our study, a large prescribed burn (5188 ha) occurred in 2003 in the management zone.

To assess if differences in beetle distribution between zones were due to active management, we needed to determine: (1) the amount of habitat in the management and monitoring zones, (2) the percent of occupied habitat, and (3) how the area affected by mountain pine beetles had changed since 1997.

2.2. Habitat

The principal source of data on the distribution and amount of habitat available for mountain pine beetles came from the Banff–Jasper Ecological Land Classification (ELC) (Holland and Coen, 1982). The ELC was conducted from 1974 to 1982 for Banff where 1900 plots were located in a stratified random design. For a subset of plots, one or two of the largest trees were cored at breast height to determine stand age (n = 587 plots). Plots were classified into vegetation types (n = 94) and a map of vegetation polygons was generated from the plot data and air photo interpretation (scale = 1:40,000 or 1:50,000).

We quantified the amount of available habitat in the different zones by first determining which ELC types were used by mountain pine beetles in the recent past (Canadian Forest Service aerial surveys 1999–2003), and then determining how much area these types covered in each zone. Most (98.5%) of the trees colonized by mountain pine beetles within the study area occurred in 11 of the 94 vegetation types. Table 3 ranks these vegetation types according to the proportion of polygons occupied by mountain pine beetles from 1998 to 2004. It is interesting to note that many of these vegetation types are not dominated by lodgepole pine, but all must contain some lodgepole pine as it is the primary host within the study area (whitebark pine, *Pinus albicaulis*, and limber pine, *Pinus flexilis* are rare). Percent occupancy is to some degree indicative of the mountain pine beetle's vegetation 'preference'. Because mountain pine beetles primarily colonize older trees (Safranyik and Carroll, 2006), the most vulnerable areas to colonization by mountain pine beetles

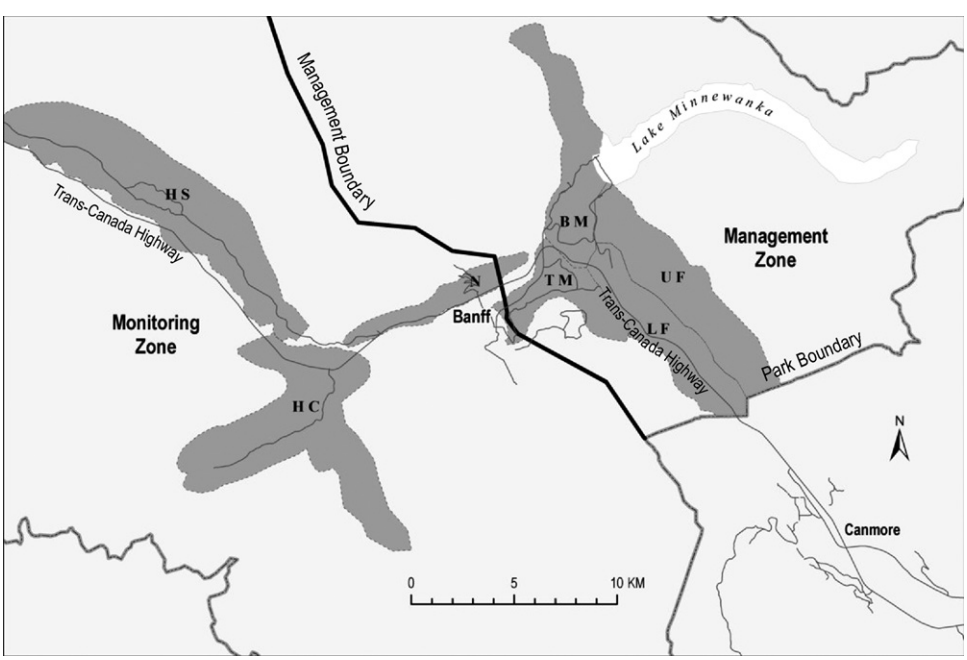

Fig. 2. Sub-zones where mountain pine beetle outbreaks occurred during the 1940s defined by Hopping and Mathers (1945) (HS, HC, N, and TM), and areas where outbreaks have occurred since 1997 (BM, LF, and UF). Heavy black line delineates the monitoring (west) and management (east) zones. Sub-zone names: BM = Bumpy Meadows, UF = Upper Fairholme, LF = Lower Fairholme, TM = Tunnel Mountain, N = Norquay, HC = Healy Creek, and HS = Hilsdale.

Table 3
Vegetation types occupied by mountain pine beetle in the Banff National Park study area since 1998, listed in order of decreasing occupancy

Vegetation type	Genera	Common names	% *P.c.*	N_{poly}	N_{MPB}	f
C11	*Pinus contorta-Picea* spp./*Hylocomium splendens*	Lodgepole pine/feathermoss	28.1	6	5	0.83
C1	*Pseudotsuga menziesii/Elymus innovatus*	Douglas fir/hairy wild rye	2.0	13	8	0.62
C16	*Populus tremuloides/Elymus innovatus-Lathyrus ochroleucus*	Aspen/hairy wild rye-peavine	0	18	10	0.56
C6	*Pinus contorta/Sheperdia canadensis/Aster conspicuous*	Lodgepole pine/buffaloberry/showy aster	42.1	54	29	0.54
C5	*Picea glauca-Pseudotsuga menziesii/Hylocomium splendens*	White spruce-Douglas fir/feathermoss	3.1	15	8	0.53
C19	*Pinus contorta/Sheperdia canadensis/Linnaea borealis*	Lodgepole pine/buffaloberry/twinflower	39.6	33	8	0.24
C3	*Pinus contorta/Juniperus communis/Arctostaphylos uva-ursi*	Lodgepole pine/juniper/bearberry	27.2	15	8	0.24
C18	*Pinus contorta/Shepherdia canadensis/Vaccinium scoparium*	Lodgepole pine/buffaloberry/grouseberry	34.5	100	18	0.18
C20	*Pinus contorta/Menziesia glabella/Vaccinium scoparium*	Lodgepole pine/false azalea/grouseberry	31.0	29	3	0.10
C13	*Picea engelmanii-Abies lasiocarpa/Hylocomium splendens*	Engelmann spruce-subalpine fir/feathermoss	1.1	100	7	0.07
O4	*Pinus albicaulis-Pinus contorta*	Engelmann spruce-subalpine fir-whitebark pine-lodgepole pine	0.1	163	9	0.06

% *P.c.* = the mean percent cover of *Pinus contorta* (lodgepole pine) from 10 plots listed in the Bannf–Jasper Ecological Land Classification (1982). N_{poly} = the number of polygons of vegetation type within the study area. N_{MPB} = the number of polygons of vegetation type with mountain pine beetles within the study area. $f = N_{MPB}/N_{poly}$. Polygons were not discriminated by the estimated stand age.

are older (>60 years) stands of these vegetation types. The distribution of these vegetation types shaded coarsely by the degree of occupancy or 'preference' is shown in Fig. 1. The aging data from the 587 plots in Banff indicates there are many old stands within the park including stands of the 11 vegetation types used by mountain pine beetles, but these are also intermixed with younger stands. It should be noted, that we focus on habitat amount and not on the spatial arrangement of habitat which could affect beetle movement and potentially the dynamics of an outbreak. Our analyses only consider stands greater than 60 years old.

To quantify changes in the spatial extent of areas affected by mountain pine beetles, we divided the study area into a 500 m × 500 m grid (4801 cells of 25 ha, 3448 cells with habitat). This scale was chosen because mark-recapture studies show that 99.7% of mountain pine beetles fly less than 250 m (Safranyik et al., 1992). Therefore, any tree within a cell potentially could have been colonized by mountain pine beetles within one or two generations. We classified a cell as 'habitat' if it contained any of the 11 vegetation types preferred by mountain pine beetles. The number and distribution of habitat cells were determined for each year of the study, accounting for aging or loss due to fire.

2.3. Abundance and distribution of mountain pine beetles

Data on trees colonized by mountain pine beetles were obtained in two ways: aerial surveys conducted by the Canadian Forest Service (recording 'red attack' trees) and trees cut by Parks Canada personnel (referred to as 'green attack' trees). Green attack trees were colonized by mountain pine beetles in the year they were cut and were identified by ground surveys, whereas red attack trees were colonized by mountain pine beetles in the previous year (so the needles had turned red by the time of the survey) and were recorded by aerial surveys. Aerial survey data were available from 1998 to 2005 and tree cutting data were available from the creation of the mountain pine beetle management zone in 2001 to 2005. Consequently, all data on mountain pine beetle distribution before 2001 were from aerial surveys. We know beetles were in Banff in 1997, but no systematic survey was completed, so we assumed the area colonized in 1997 was 0.5 ha which corresponds to 11–50 trees (double the detection limits of an aerial survey). In addition, the 2004 aerial survey did not cover the area burned in the management zone in 2003 because of the discoloration of trees which could have been caused by either fire or beetles (Unger, 2004). Therefore, estimates of area affected in 2003 are biased low and should be interpreted with caution.

After the creation of the two zones, the beetle data in the management zone were a mix of aerial survey and tree cutting data, whereas in the monitoring zone the data only came from aerial surveys. These data sources presented some problems. Not only were the two zones treated differently, the data telling us the response of mountain pine beetle populations were also different. Therefore, there was some confounding of the signal of interest (mountain pine beetles) with the methods in which the data were collected. Recognizing the difficulty in the data and its effect on interpreting the results, it was necessary to combine the point (tree removal) and polygon (aerial survey) datasets to estimate spatial spread of mountain pine beetles. Combining datasets puts the estimate of area affected by mountain pine beetle on the same spatial scale thereby improving the comparability between zones. That is, we recreated a pattern that satisfied the question: 'What *would* have been the distribution of red attack trees in 2005 (for example) if no green attack trees had been removed?' to infer the area colonized in 2004. To combine the data, we used the standard developed by Canadian Forest Service to estimate the area affected in small 'spot' infestations: 2–10 colonized trees were assumed to cover 0.25 ha and 11–50 trees to cover 0.5 ha (http://www.for.gov.bc.ca/hfp/health/overview/facts.htm). If we assume a single tree can be detected by an aerial survey, use 5 and 30 trees as the midpoint for the ranges, and assume a linear relationship between trees colonized and the radius of a spot, we can calculate the radius affected (r) around each colonized tree (x). The radii corresponding to 0.25 and 0.5 ha, assuming a circle, were 28.22 and 39.90 m, respectively. The slope was then calculated as (39.90–28.22)/(30–5) = 0.467 and the intercept as 25.88 m.

$$r = 0.467x + 25.88,$$

where x = number of trees colonized.

This conversion may seem large to any observer on the ground, but it really is only relevant to an observer in an airplane. We then buffered the number of colonized trees removed by Parks personnel at a given location by the equation above (in Arc GIS) to get an estimate of how much area would have been recorded in an aerial survey had the trees not been removed. Because many of these buffered locations overlap to create larger polygons, we excluded this overlap in our GIS manipulations and calculations of area affected.

2.4. Spatial contrast

Comparisons between the management and monitoring zones were done several ways. We measured the abundance of mountain pine beetles using three metrics: area colonized by mountain pine

beetles, the number of 25 ha cells with colonized trees, and the proportion of cells with available habitat that were occupied. The area colonized provides the best precision, but because of the estimation needed to combine green and red attack data, it may have some bias. The number of cells with colonized trees may compensate for this by looking simply at presence in each cell, and it also captures the spatial extent of areas affected by mountain pine beetles. The proportion of occupied cells with habitat controls for habitat availability. For each of these metrics, we also examined year-to-year changes which were, respectively, the rate of spatial spread ($\log_e$(area colonized next year/area colonized in current year)), the year-to-year change in the number of cells with colonized trees and the year-to-year change in the proportion of available habitat cells occupied by mountain pine beetles. Typically, rates of change are used to estimate population growth rate (e.g. Sibly et al., 2005), however, in our case we apply it to the area occupied and therefore it is a measure of the rate of spatial spread. To get an idea if smaller areas within a zone were acting similarly, we divided each zone into sub-regions following the areas outlined by Hopping and Mathers (1945) and while adding a few of our own (Fig. 2). It is interesting to note that the current outbreak occurred in four of the same areas as in the 1945 outbreak (Fig. 2; HS, HC, N, and TM), but also in three new areas to the east (BM, LF, and UF). The sub-zones reflected concentrations of colonized trees and broadly similar biophysical characteristics. We then compared the area colonized in these sub-zones.

It is inevitable that some colonized trees would be missed in the management zone, and the aerial survey in the following year estimated the extent to which trees were missed. Polygon size of colonized patches was divided into two size categories: less than 0.5 ha and greater than 0.5 ha. Polygons less than 0.5 ha were assumed to have 2–10 colonized trees, and polygons greater than

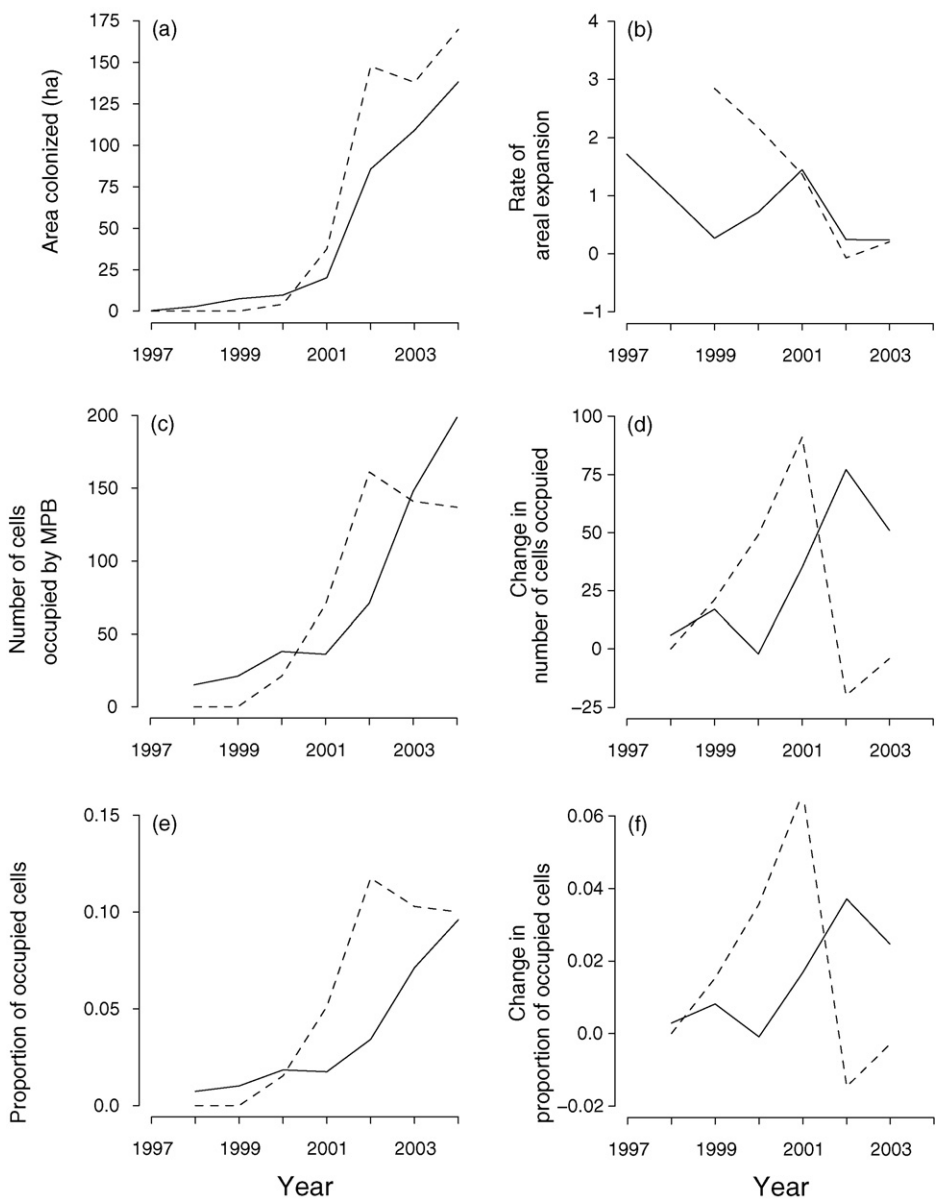

Fig. 3. Mountain pine beetle (MPB) increase and spread in the monitoring (solid line) and management (dashed line) zones. (a) Area colonized by MPB, (b) the rate of areal increase ($\log_e$(area colonized next year/area colonized in current year)) by MPB, (c) number of 25 ha cells with MPB colonized trees, (d) year-to-year change in the number of cells with MPB colonized trees, (e) proportion of available habitat cells occupied by MPB, and (f) year-to-year change in the proportion of available habitat cells occupied by MPB. Number of cells with habitat: monitoring zone, 2077 and management zone, 1371. Year indicates the year trees were colonized.

0.5 were assumed to have 11–50 colonized trees. Only a few polygons in the management zone exceeded 1 ha in size. We estimated a lower and upper limit of number of trees missed in the management zone by multiplying polygon number in each category in each year by the lower and upper bounds of the number of trees colonized per polygon.

The interpretation of these comparisons can only pertain to the study area. Without statistical replication (i.e. multiple areas that are treated similarly and are statistically independent), which would provide some estimation of process error, we are unable to state the probability of a similar pattern occurring elsewhere. This problem is typical of large-scale studies of biological patterns. We can, however, state if patterns are similar or significantly different based on sampling error. We were unable to find a published estimate of sampling error of aerial surveys derived either from multiple observers or multiple flights. Most studies compare aerial estimates with ground surveys (Harris et al., 1982; Wulder et al., 2006), which is less useful for our simple comparison between management zones. However, the BC Ministry of Forests and the Canadian Forest Service (Anonymous, 2000) state that their accuracy tolerance limits are within 30% of the original assessment (i.e. CV = 0.3), and we used this value in our comparisons. We suggest that the importance of an observed temporal or spatial pattern should be judged on both statistical and biological relevance.

3. Results

In 1997, the 11 vegetation types in which mountain pine beetles were typically found and that were more than 60 years old covered 47% (55,964 ha) of the study area (Table 2). Habitat amount increased by trees entering susceptible age classes and decreased when killed by fire. On average 334 ha burned per year in the study area since 1983, with a large burn (5188 ha) occurring in the management zone in 2003 and no fires in 10 of the years. Since 1983, habitat amount fluctuated by 5362 ha (ca. 10%) which was caused more by fire than by tree aging or removals. The management zone was 62% smaller than the monitoring zone, and contained 30% less habitat in 1997. Consequently, the percent habitat is roughly similar: habitat covered 50.2% of the management and 44.4% of the monitoring zone (Table 2). However, habitat in the management zone tended to be of higher preference by mountain pine beetles (Fig. 1). The percentage of all habitat that was composed of those vegetation types that had more than 50% polygon occupancy by mountain pine beetles was 27.8% and 10.0% for management and monitoring zones, respectively.

Overall, an estimated 601.6 ha were affected by mountain pine beetles in the study area between 1997 and 2004 (Table 2). Based on the rough estimates of stand age, density and the severity of the attack, approximately 600,000 trees were colonized in 499 cells.

The area colonized by mountain pine beetles increased rapidly in the management and monitoring zones in 2001 and 2002, with the area colonized in the management zone exceeding the monitoring zone after 2000 (Fig. 3a). In 2004, 169.8 ha were colonized in the management zone and 138.1 ha in the monitoring zone. These areas are not significantly different if a CV of 0.3 is assumed (95% CI: 36–240 ha management zone, 55–221 ha monitoring zone), or even if a CV of 0.1 is assumed (95% CI: 104–172 ha management zone, 110–166 ha monitoring zone). Re-scaling these estimates by the proportion of habitat in each area makes the estimates more similar.

The rate of spread of area colonized initially was high in the monitoring zone (Fig. 3b), but the exact value depended on the colonized area assumed in 1997 (0.5 ha). In the management zone, the rate of spread initially was high but declined rapidly and was negative (indicating contraction) in 2002. The area colonized in the management zone in 2003, and consequently, the rate of spatial spread in 2002 is biased low because the aerial survey did not include the area burned by the prescribed fire. This area was ground surveyed, however, and the 2003 and 2004 tree removals should include baited trees and isolated pockets of colonized trees from this area. By 2003, mountain pine beetles were expanding at about 1.23 ha per year in both the management and monitoring zone (Fig. 3b).

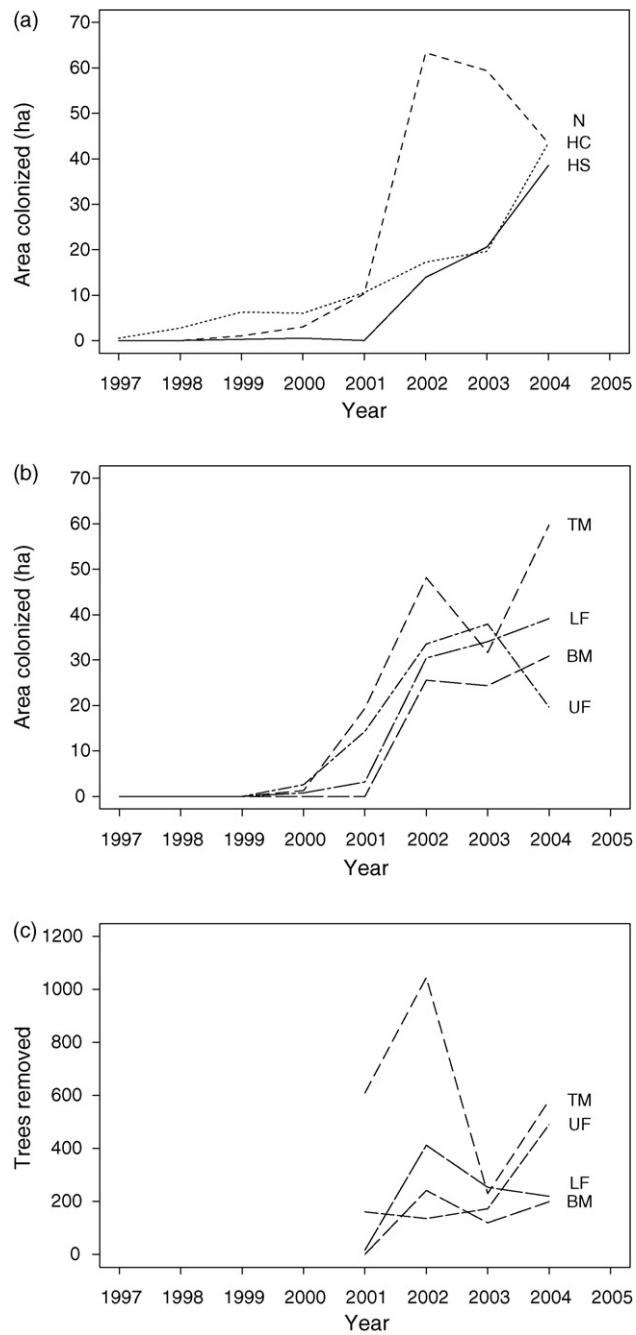

Fig. 4. Area colonized by MPB within sub-zones estimated from tree removal and aerial survey data in the monitoring zone (a) and management zone (b), and the number of trees cut and removed within sub-zones in the management zone, (c). Sub-zone names: BM = Bumpy Meadows, UF = Upper Fairholme, LF = Lower Fairholme, TM = Tunnel Mountain, N = Norquay, HC = Healy Creek, and HS = Hilsdale. See Fig. 2 for the location of sub-zones. Year indicates the year trees were colonized or removed.

Table 4
The effect of the $\log_e$ area colonized by mountain pine beetles on the rate of spatial spread within management zone

Parameter	Analysis of variance				Parameter estimate		
	d.f.	Sum of squares	*F* value	*p* value	Coefficient	S.E.	*p* value
Zone	1	0.73	3.62	0.09	1.14	0.42	0.03
$\log_e$(area)	1	6.27	30.98	<0.001	−0.23	0.10	0.05
Zone*$\log_e$(area)	1	0.54	2.69	0.14	−0.21	0.13	0.14
Residual	8	1.62					

Model intercept = 1.36; S.E = 0.29.

There were 3448 cells (500 m × 500 m) containing some habitat, with more cells in the monitoring (2077) than management zone (1371). Trees colonized by mountain pine beetles occurred in 499 cells between 1997 and 2004. Beetles were initially concentrated in the western part of the study area around Healy Creek (Fig. 2). The extent of the mountain pine beetle infestation increased only slightly in 1999, but rapid increases were observed in 2000 and by 2001 mountain pine beetles colonized Mt. Norquay and in 2003 the lower Fairholme bench. The number of cells occupied by mountain pine beetles (Fig. 3c) shows a somewhat different pattern than a direct measure of areal extent (Fig. 3a). Beetles were colonizing new cells at a faster rate in the management than monitoring zone (Fig. 3d), but this trend was stopped in 2002 and 2003. This change in colonization rates could have been a direct effect of the bait and removal program, to the removal of habitat of high preference to mountain pine beetles by a prescribed burn (where lodgepole pine was killed), or because the burned areas were not fully surveyed (Unger, 2004). This resulted in a marked reduction in the magnitude of year-to-year increases in the number of cells occupied (Fig. 3d). Accounting for the number of cells with habitat magnified the difference between the trends in habitat occupancy in each zone (Fig. 3e), but not the rate in which they were occupied (Fig. 3f). The proportion of habitat cells occupied continued to increase in the monitoring zone, whereas after 1 year (by 2002) of the bait, fell and burn policy the proportion of occupied habitat in the management zone remained steady, also resulting in a strong decrease in the year-to-year change in the proportion of habitat cells occupied in the management area compared to the monitoring area (Fig. 3f).

Not all sub-zones within the management or monitoring zones increased at the same rate and two sub-zones showed decreasing trends (Norquay and Upper Fairholme) suggesting that there was spatial variability in the increase and spread of mountain pine beetles (Fig. 4). Notably, the Norquay site was within the monitoring zone, and its decrease may be attributed to local depletion of trees as the area affected in 2002 alone represented 7.5% of that area. Of the sub-zones in the management area (Fig. 4b), the Fairholme sub-zones were impacted by the prescribed fire in 2003 while the other two were not. The fire affected 781 ha (38.6%) of Lower and 2186 ha (80.7%) of Upper Fairholme, which showed the sharpest decrease in area affected.

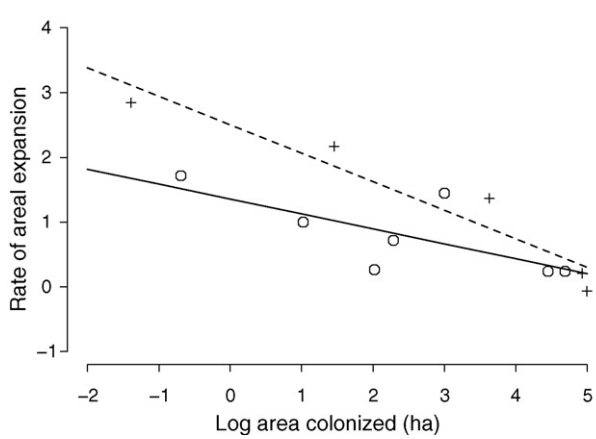

Fig. 5. The observed rate of spatial expansion for the monitoring (circles) and management (pluses) zones. Lines are statistical fits to the monitoring (solid) and management (dashed) zones. Statistical results reported in Table 4.

The number of trees removed did not necessarily predict the extent of the decrease. Between 2001 and 2004, 1369 baits were deployed and a total of 4883 lodgepole pine trees that were colonized by mountain pine beetles were cut and removed from the management zone (Table 2). The greatest number of trees were removed from Tunnel Mountain (2464 trees) and the least from Bumpy Meadows (560 trees) (Table 2, Fig. 4c). In only two cases were removals coincident with decreases in area colonized. Removal of colonized trees from Tunnel Mountain in 2002 was coincident with a decrease in area colonized in 2003, and the removal of trees from Upper Fairholme in 2003 was coincident with a decrease in area colonized in 2004. In all other areas and years the area colonized either increased or remained stable despite the removals.

The number of trees colonized but not cut and removed was difficult to estimate because it is a function of the area estimated from aerial surveys in the following year and the assumed number of trees affected per spot. Despite extensive effort, somewhere between 1222 and 5760 colonized trees were not removed in 2004. This is based on the area of red attack in 2005 assuming 2–10 trees per ha for spots less than 5 ha (n = 226 polygons) and 11–50 trees per ha for spots greater than 5 ha (n = 70 polygons). Given that 1490 trees were removed in 2004, somewhere between 45 and 79% of the trees colonized in the management zone in 2004 were missed. This wide range shows the difficulty in estimating the number of trees missed without on-the-ground counting of red attack trees. This estimate is biased high, however, because aerial survey data cannot differentiate mountain pine beetle colonization from other sources of mortality (*Ips* colonization, drought).

The larger the area colonized by mountain pine beetles the slower the rate of areal increase, and the slope of this relationship did not detectably depend on control efforts (zone effect is non-significant, Table 4, Fig. 5).

4. Discussion

Mountain pine beetles have been increasing in the study area since 1997, but the rate of increase has been slowing. Using area colonized, there was no detectable effect of management at the scale of the zone (Table 4, Figs. 3a, 3b and 5). After 4 years, the control measures did not significantly affect the area colonized by mountain pine beetles, and in 2004, mountain pine beetles were still expanding in both zones, although at a slow rate (1.23 ha per year). This conclusion is conservative, because the prescribed fire in the management zone in 2003 may have artificially reduced the estimated area of beetle-killed trees. At the sub-zone scale, the area affected within the management zone tended to have stabilized

more than those in the monitoring zone (Fig. 4b vs. Fig. 4a), but overall the variation among sub-zones obscured any clear differences among zones.

Using 25 ha cells (Fig. 3c–f), it appears that the spatial extent and the rate at which new cells were colonized was much reduced in the management zone relative to the monitoring zone. In this sense, the spread (in terms of cells colonized) was reduced in the management zone. These results appear to contradict those for areal extent just discussed, but they actually suggest complex spatial dynamics occurring on this landscape. Beetles were less successful colonizing new areas separated by 500 m or more in the management zone, indicating some effectiveness in the bait and removal program, or the prescribed fire, in slowing their large-scale spatial spread. It is possible that the reduction in the number of cells with mountain pine beetles is due to a direct effect of the removal of susceptible lodgepole pine or an indirect effect by altering movement patterns, or both. It appears that short distance dispersal remained successful as indicated by the continued increase in area colonized.

A decreasing rate of area colonized (Fig. 3b) along with an 8% difference in occupied habitat cells in 2003 (Fig. 3e) led us to conclude that the bait, cut and removal management strategy in the management zone was effective in *slowing* the spread of mountain pine beetles. It should be noted that the Park's objective was not to stop the spread of mountain pine beetles in management zone, but rather to slow the spread to the region east of the park boundary. This objective was not specifically addressed in this study. Our conclusions about the effectiveness of management actions must be tempered by the limitations of this study. A spatial comparison without replication and a disparity in how data were collected in each zone preclude strong conclusions about the effectiveness of management. The area colonized in the management zone in 2004 was not statistically different from the monitoring zone based on sampling error. The rate of spatial spread depended on the area colonized the previous year and not on management zone (Table 4, Fig. 5). Therefore, we conclude that control efforts had limited success in slowing the increase of mountain pine beetles within Banff National Park. It is possible, though, that the management zone acted as a sink and prevented mountain pine beetle colonization to the east, but this scenario was not tested.

Underlying environmental differences between the two zones, such as the prevalence of south facing slopes or precipitation, could have led to early differences in the rates of spread in the two zones. Greater amounts of habitat in the monitoring zone should have generated faster rates of expansion than in the management zone, but more 'preferred' habitat possibly of higher quality in the management zone may have countered differences in habitat amount. We observed some spatial variability in the area colonized among sub-areas within each zone (Fig. 4). Consequently, the differences observed in Fig. 3 could be insignificant when compared to background variability.

The area with the most trees removed (Tunnel Mountain) also showed the largest decrease in the area colonized by beetles in the following year, but this trend was not sustained and the area colonized by beetles increased in 2004 to its highest levels. This trend implies that any benefits of tree removal on Tunnel Mountain were short-term and local. Local decreases in abundance were possibly overshadowed by a regional increase in mountain pine beetle abundance and the movement of beetles back into the area. Changes in the greater bark beetle community may have also caused increased tree mortality (Logan et al., 1998). Ground surveys indicated that much of the tree mortality on Tunnel Mountain in 2004 was due to *Ips* colonization (I. Pengelly, Banff National Park, personal communication).

In order for management measures to stop the spread of mountain pine beetles, it is necessary to observe either range contraction or halted expansion. Carroll et al. (2006) showed how the success of management depends on population rates of increase, the proportion of trees treated, and the size of the outbreak. Mortality needs to be >97.5% for a mountain pine beetle population to decline (Safranyik and Carroll, 2006), and the consequences of missing a few trees could be large. It may be useful to add a spatial dimension to the mortality factor and the probability of successful control discussed in Carroll et al. (2006). That is, mortality needs to be >97.5% over some defined area. If we assume that mortalities are additive and that beetles from trees, say within 25 ha, can still interact with each other then we can show how the success of control measures depends on the rate in which colonized trees are missed and on brood survival.

$$N_t = (N_0 ESP)^t,$$

where N_0 = the number of beetles within 25 ha, E = the number of eggs per beetle, S = brood survival, P = the proportion of trees missed in 25 ha, and t is the number of years. If each beetle can lay 60 eggs (Elkin and Reid, 2005), only 3% of colonized trees were missed in a 25 ha area, and there was 100% survival (although unlikely) then, theoretically, the population could increase. If 30% were missed and there was 10% survival then the population could also increase. This occurs because mortality over 25 ha remains less than 97.5%. So, successful control of mountain pine beetles depends on removing a high proportion of colonized trees and low brood survival rates in missed trees. Because survival rates are variable and are thought to be weather dependent, the consequences of missing trees changes from year to year (Reid, 1962, 1963; Cole, 1981; Amman, 1984). In addition, mountain pine beetles are thought to be resource limited, at least locally (Trzcinski and Reid, in press). Local reductions in abundance could be offset by higher fecundity, making control efforts even more difficult. We estimated that somewhere between 45 and 79% of the trees colonized in the management zone in 2004 were missed, but this is only 0.7–3.7 trees per 1000 ha of habitat! This is a low miss rate when considering the remoteness of the area and difficulty of the terrain, but missing a few hundred trees may allow localized outbreaks to continue. Ultimately, the success of management depends on removing a high proportion of colonized trees and on limiting the dispersal of beetles into a management area. Even if 100% of colonized trees are removed an influx of beetles from other areas may sustain an outbreak.

While conclusions about the effectiveness of mountain pine beetle management in Banff National Park must be tentative, it appears that management measures have slowed the rate of spread, but not stopped it. Our conclusion differs from Clarke and Billings' (2003) analysis of similar data for southern pine beetle, *Dendroctonus frontalis*, where they concluded that control was very successful. It is difficult to directly compare the two studies because of differences in how the data are reported. However, if management is more successful for southern pine beetles than we have observed for mountain pine beetles, it could be that the two species differ in the discreteness of patches of colonized trees (Fettig et al., 2006), making it easier to remove most of the colonized trees for southern pine beetles. Bentz and Munson (2000) also reported high success in reducing spruce beetles, *Dendroctonus rufipennis,* over a 500 ha area by attracting beetles to baited traps and felled trees along with selective removal of infested trees, whereas an adjacent unmanaged areas had increases in the spruce beetle populations. Here the success may be linked to the greater propensity of spruce beetles to attack downed rather than standing trees, where downed trees are easier to identify and remove (Bentz and Munson, 2000). Further studies

at the regional scales such as these and ours will be useful for refining management tactics for different bark beetle species (Fettig et al., 2007).

Our study constitutes a rare test of the effectiveness of direct control, with many favorable components including two management treatments within one jurisdiction, available vegetation maps, and annual surveys of the areas affected by mountain pine beetles. We used these to develop a procedure to assess management. We found that even with these favorable circumstances, we were unable to be strongly conclusive about management effectiveness, likely due in part to data limitations. The best way to improve the evaluation of management measures is to reduce the differences in data collection by conducting ground surveys where both red and green attack trees are counted in areas where tree removal is occurring, by using statistically replicated management areas, and not occurring and by spatial population modelling. We suggest that the greatest improvement in predicting the pattern of future spread would be gained by conducting a comprehensive survey of vegetation age structure and spatial variation in tree growth rates (habitat modelling). In the current outbreak of mountain pine beetles, governments in Canada and the United States have directed hundreds of millions of dollars to research and management of mountain pine beetles, and similar large investments have been made in previous outbreaks. However, critical analyses of the success of management practices are almost non-existent. Given the difficulty of conducting controlled experiments in this system, robust conclusions will need to be obtained through repeated studies such as the current one, necessitating investment in data gathering throughout the management action.

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